

TAYLORVILLE, IL, USA

WMO: 744662

Lat: 39.534N

Lon: 89.328W

Elev: 190

StdP: 99.07

Time Zone: -6.00 (NAC)

Period: 02-19

WBAN: 63817

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.7	-13.1	-21.2	0.6	-15.1	-18.0	0.8	-12.4	12.6	-0.8	11.4	-2.1	3.3	320	0.499

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.3	34.1	24.4	32.8	24.0	31.8	23.6	26.7	31.8	25.7	30.8	24.9	29.7	3.2	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.2	20.9	29.9	24.1	19.4	28.3	23.3	18.5	27.2	84.6	31.1	80.7	31.1	76.4	30.2	30.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.9	9.3	8.4	DB	-20.8	36.1	3.4	3.1	-23.2	38.3	-25.2	40.2	-27.1	41.9	-29.6	44.2	(4)
(5)			WB	-21.1	28.2	3.3	0.8	-23.6	28.7	-25.5	29.2	-27.4	29.6	-29.8	30.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	12.2	-2.3	-0.8	6.4	12.5	18.4	23.0	24.3	23.3	20.4	13.3	6.2	0.6	(6)
(7)		DBStd	10.74	6.58	6.35	6.35	5.37	4.83	3.21	3.27	3.29	4.19	5.35	5.56	5.45	(7)
(8)		HDD10.0	1333	383	307	148	34	2	0	0	0	0	26	140	293	(8)
(9)		HDD18.3	2918	639	537	371	184	62	4	2	3	26	175	366	549	(9)
(10)		CDD10.0	2122	2	3	38	110	262	389	442	412	311	127	24	2	(10)
(11)		CDD18.3	666	0	0	2	11	63	143	185	157	87	18	1	0	(11)
(12)		CDH23.3	6368	0	0	19	115	551	1344	1851	1455	863	168	3	0	(12)
(13)	CDH26.7	2461	0	0	1	26	166	516	783	585	338	47	0	0	(13)	
(14)	Wind	WSAvg	3.7	4.7	4.4	4.6	4.8	3.9	3.0	2.2	2.1	2.5	3.5	4.3	4.4	(14)
(15)	Precipitation	PrecAvg	959	61	44	66	97	120	112	85	84	71	86	82	55	(15)
(16)		PrecMax	1226	168	130	128	235	271	213	195	134	230	168	194	166	(16)
(17)		PrecMin	764	13	13	24	40	33	57	35	20	3	24	9	17	(17)
(18)		PrecStd	133	35	29	29	47	69	45	42	25	50	40	44	35	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.2	18.0	26.2	29.2	32.7	34.1	37.8	37.2	35.9	31.0	23.1	17.1	(19)
(20)			MCWB	13.9	13.6	17.8	17.6	22.6	24.2	25.4	25.5	22.7	21.4	16.3	14.9	(20)
(21)		2%	DB	12.7	15.0	22.4	27.0	30.2	32.8	34.1	33.9	32.4	27.7	19.7	13.6	(21)
(22)			MCWB	10.9	12.3	16.3	16.9	21.4	23.7	25.4	24.3	22.3	19.1	15.2	11.6	(22)
(23)		5%	DB	9.8	11.7	18.9	23.9	28.2	31.7	32.6	32.3	31.0	24.9	17.3	11.3	(23)
(24)			MCWB	8.1	9.2	14.5	16.0	20.1	23.3	25.1	24.1	22.1	17.9	13.5	9.6	(24)
(25)	10%	DB	6.4	8.0	16.4	22.1	27.0	29.8	31.1	30.0	28.0	22.3	14.9	8.0	(25)	
(26)		MCWB	4.5	5.8	12.7	15.1	19.4	22.5	24.5	23.3	20.3	16.7	11.7	6.4	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.6	15.1	19.0	20.3	24.3	26.9	28.1	27.7	25.3	22.7	18.7	15.0	(27)
(28)			MCDWB	16.0	17.3	24.1	26.3	30.4	31.6	34.0	33.4	31.6	28.2	21.9	16.7	(28)
(29)		2%	WB	11.5	12.7	17.0	18.7	22.8	25.6	27.0	26.4	24.1	21.2	16.3	12.4	(29)
(30)			MCDWB	12.6	14.3	20.8	22.8	28.2	30.7	32.3	31.5	29.1	25.4	18.7	13.3	(30)
(31)		5%	WB	8.1	9.1	15.3	17.4	21.7	24.5	26.1	25.3	23.2	19.6	14.3	9.4	(31)
(32)			MCDWB	9.1	11.3	18.5	21.7	26.4	29.6	31.0	30.5	27.6	22.7	16.7	10.9	(32)
(33)	10%	WB	4.7	6.1	13.2	15.9	20.5	23.5	25.2	24.3	22.1	17.8	12.0	7.0	(33)	
(34)		MCDWB	6.7	8.2	16.0	20.6	24.9	28.2	29.9	28.4	26.4	21.3	14.4	8.0	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.3	8.9	10.1	11.6	11.1	11.4	11.3	11.6	13.2	11.9	9.6	7.5	(35)
(36)			MCDBR	12.2	12.9	13.5	14.6	13.2	13.2	12.8	13.7	14.6	14.4	12.9	10.8	(36)
(37)			MCWBR	10.3	10.2	8.5	7.5	5.8	5.5	5.2	5.7	5.5	6.9	8.9	9.1	(37)
(38)		5% WB	MCDBR	11.5	12.3	11.8	12.5	11.2	11.3	11.1	11.3	11.8	11.9	11.5	10.1	(38)
(39)			MCWBR	10.4	10.3	7.9	8.1	5.7	5.5	5.5	5.4	6.7	9.3	9.3	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.302	0.313	0.349	0.391	0.431	0.429	0.460	0.449	0.407	0.357	0.310	0.297	(40)	
(41)		taud	2.453	2.411	2.386	2.269	2.214	2.254	2.156	2.176	2.275	2.421	2.505	2.483	(41)	
(42)		Ebn at Noon	871	908	906	884	852	851	820	819	832	845	855	849	(42)	
(43)		Edh at Noon	80	95	109	131	141	136	149	142	121	94	75	72	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.88	2.71	3.67	4.86	5.50	6.32	6.25	5.55	4.69	3.28	2.19	1.58	(44)	
(45)		RadStd	0.22	0.25	0.28	0.30	0.45	0.47	0.41	0.36	0.32	0.36	0.22	0.20	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (38 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+107	+78

Nomenclature: See separate page